Paper 2

Mathematics 4052/02
October/November 2024
2 hours 15 minutes

Answer all the questions.

1. **(a)** Find all the integers that satisfy the inequality $-4 \leqslant 3 + 2x < 5$.

Answer .... [3]

(b) Simplify $\dfrac{20p^3q}{7r^2} \div \dfrac{8p^2r}{21q^4}$.

Answer .... [2]

(c) Simplify $\left(\dfrac{a^9}{64b^{12}}\right)^{-\frac{2}{3}}$.

Answer .... [2]

(d) Write as a single fraction in its simplest form $\dfrac{x}{x+2} + \dfrac{3}{4-x} - 1$.

Answer .... [4]

2. **(a)** A cinema has three halls.

Each hall has standard seats and premier seats.

Hall 1 has 120 standard seats and 40 premier seats.

Hall 2 has 90 standard seats and 30 premier seats.

Hall 3 has 50 standard seats and 20 premier seats.

This information can be represented by the matrix $\mathbf{C} = \begin{pmatrix} 120 & 40 \\ 90 & 30 \\ 50 & 20 \end{pmatrix}$.

The ticket price for a standard seat is \$12.

The ticket price for a premier seat is \$15.

(i) Represent the ticket prices in a 2×1 column matrix **T**.

Answer **T** = [1]

(ii) Evaluate the matrix $\mathbf{M} = \mathbf{CT}$.

Answer **M** = [2]

(iii) The cinema expects to sell $\frac{3}{5}$ of each ticket type in all halls for a Monday night show.

The cinema expects to sell $\frac{9}{10}$ of each ticket type in all halls for a Saturday night show.

There is one film shown in each hall every night.

Calculate the difference between the expected takings for all halls on a Monday night and for all halls on a Saturday night.

Answer \$ .... [2]

(b) In the first week of the month, 5075 tickets were sold.

In the second week of the month, sales decreased by 56%.

Calculate the number of tickets sold in the second week of the month.

Answer .... [2]

(c) Each hall shows one film in the afternoon, one film in the evening and one film at night.

On a Sunday, tickets were sold in the following ratios:

afternoon tickets : evening tickets = 3 : 8

evening tickets : night tickets = 6 : 11

Find the ratio afternoon tickets : night tickets sold that Sunday.

Give your answer in its simplest form.

Answer .... : .... [2]

3. **(a)** The equation of line L is $4y = 3x - 2$.

The equation of line M is $2y = kx + 13$, where k is a constant.

Line L and line M intersect at the point $(-4, p)$, where p is a constant.

Find the value of k and the value of p.

Answer $k =$ ....

$p =$ .... [3]

(b) X is the point $(-3, 7)$.

$\overrightarrow{XY} = \begin{pmatrix} 5 \\ -2 \end{pmatrix}$.

Find the equation of the line XY.

Answer .... [3]

(c) A is the point $(8, a)$, where $a < 0$.

B is the point $(15, 11)$.

The length of line AB is 25 units.

Find the value of a.

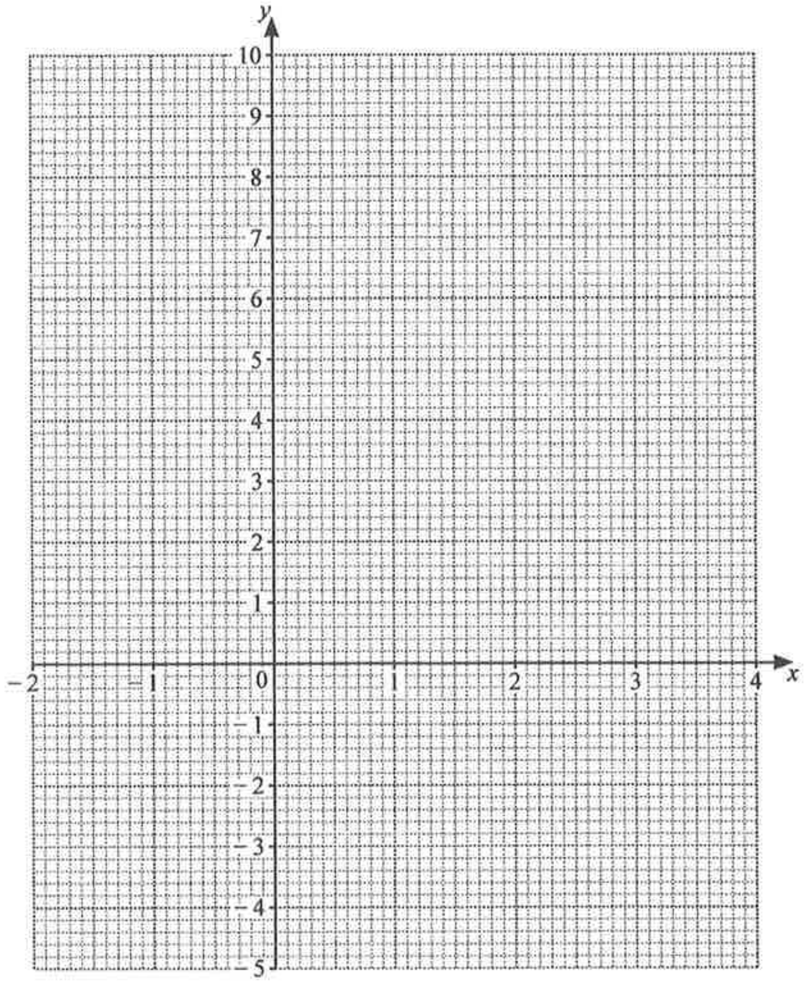
Answer $a =$ .... [3]

4. (a) (i) Complete the table of values for $y = 2x^2 - \frac{x^3}{2} - 4$.

x	-2	-1	0	1	2	3	4
y		-1.5	-4	-2.5	0	0.5	-4

[1]

(ii) On the grid, draw the graph of $y = 2x^2 - \frac{x^3}{2} - 4$ for $-2 \leqslant x \leqslant 4$.



[3]

(iii) Use your graph to explain why the equation $2x^2 - \frac{x^3}{2} - 4 = x$ has only one solution.

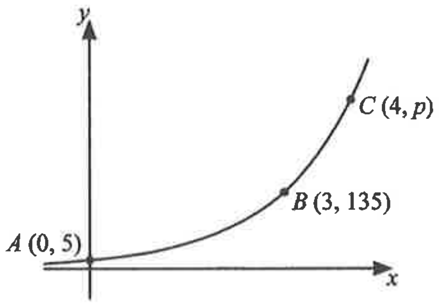
....

.... [1]

(iv) By drawing a suitable straight line on the grid, solve the equation $4x^2 - x^3 - 4 = 0$.

Answer .... [3]

(b)



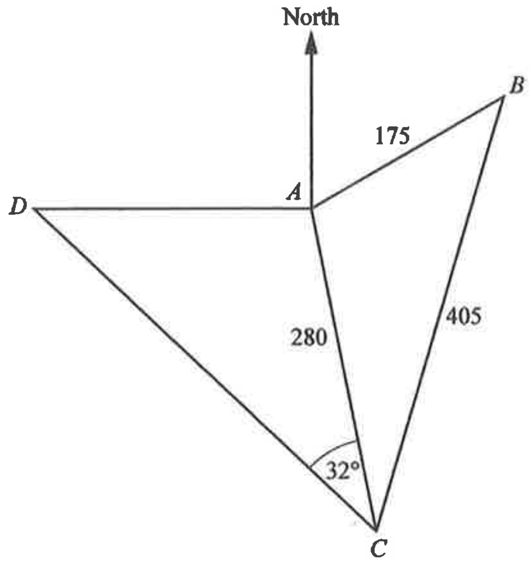
The sketch shows the graph of $y = ka^x$, where $a > 0$.

The graph passes through the points $A(0, 5)$, $B(3, 135)$ and $C(4, p)$.

Find the value of p.

Answer $p =$ .... [3]

5.



Points A, B, C and D are on horizontal ground.

$AB = 175$ m, $AC = 280$ m, $BC = 405$ m and angle $ACD = 32°$.

B is on a bearing of 048° from A.

D is due west of A.

(a) Show that angle $BAC = 124.1°$, correct to one decimal place.

Answer

[3]

(b) Calculate AD.

Answer .... m [3]

(c) Point A is the base of a vertical mast AT.
The angle of elevation of T from C is 34°.
Calculate the angle of elevation of T from B.

Answer .... [3]

6. Nadia drives the first 40 km of a journey on an expressway and the remaining 20 km on local roads.
She drives on the expressway at an average speed of x km/h.
Her average speed on the local roads is 30 km/h slower than her average speed on the expressway.
The journey takes Nadia a total of 50 minutes.

(a) Show that $x^2 - 102x + 1440 = 0$.

Answer

[5]

(b) Solve the equation $x^2 - 102x + 1440 = 0$.
Give your solutions correct to two decimal places.

Answer $x =$ .... or $x =$ .... [3]

(c) Explain why one of the solutions in **part (b)** must be rejected.

....

....[1]

(d) Calculate the difference between the times Nadia spent driving on the expressway and driving on local roads.
Give your answer in minutes and seconds, correct to the nearest second.

Answer .... minutes .... seconds [2]

7. (a) There are 120 apples in a box.

The table shows the distribution of the masses of these apples.

Mass (m g)	$60 < m \leqslant 80$	$80 < m \leqslant 100$	$100 < m \leqslant 120$	$120 < m \leqslant 140$	$140 < m \leqslant 160$
Frequency	8	21	27	46	18

(i) Find the interval that contains the median mass.

Answer .... $< m \leqslant$ .... [1]

(ii) Calculate an estimate of the mean mass.

Answer .... g [1]

(iii) Calculate an estimate of the standard deviation of the masses.

Answer .... g [1]

(iv) Another 80 apples are added to the box.

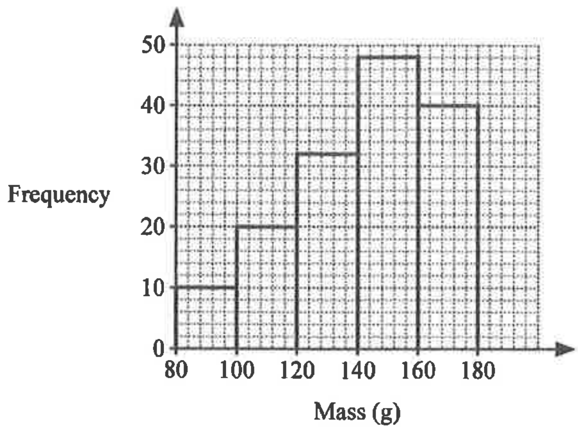
The estimated mean mass of the apples in the box is now 110 g.

Explain what this tells you about the masses of the apples added to the box.

....

.... [1]

(b) The histogram shows the distribution of the masses of 150 tomatoes.



(i) Explain why the range of masses of tomatoes may not be 100g.

....

.... [1]

(ii) One of the tomatoes is selected at random.

Find the probability that it has a mass between 120 g and 160 g.

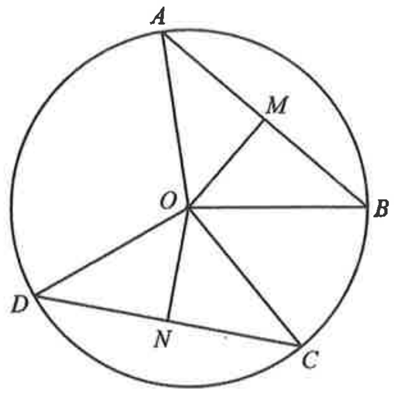
Answer .... [1]

(iii) Three of the 150 tomatoes are selected at random without replacement.

Find the probability that two have a mass less than 120 g and one has a mass greater than 160 g.

Answer .... [3]

8. (a)



A, B, C and D are points on the circle, centre O.

M is the midpoint of AB and N is the midpoint of CD.

$AB = CD$.

(i) Show that triangle BMO is congruent to triangle CNO.

Give a reason for each statement you make.

....

....

....

....

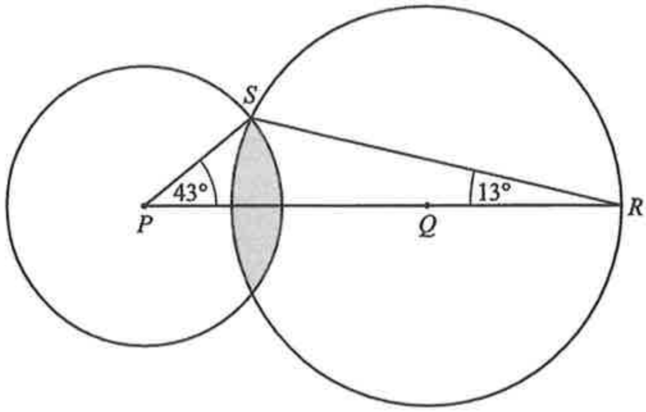
.... [3]

(ii) The radius of the circle is 4 cm and angle $COD = 1.8$ radians.

Calculate the length of the major arc AB.

Answer .... cm [2]

(b)



The diagram shows two circles, centres P and Q.

S is a point of intersection of the circles and PQR is a straight line.

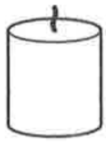
Angle $RPS = 43°$ and angle $PRS = 13°$.

The radius of the larger circle is 7 cm and the radius of the smaller circle is 4.5 cm.

Calculate the shaded area.

Answer .... cm^2 [5]

9. Wei starts a small business making candles.
He makes the candles by pouring wax and fragrance oil into a mould.
He uses a length of string as the wick.
The tables below show information Wei uses to calculate the materials he needs.



Calculating materials for a candle

- Total mass of candle in grams = 0.765 × volume of mould in cubic centimetres
- Total mass of candle = mass of wax + mass of fragrance oil
- Mass of fragrance oil = 8% of mass of wax
- 1 ml of fragrance oil has a mass of 1 gram
- The wick is 5 mm longer than the height of the candle
- All masses are calculated correct to the nearest gram

Candle supplies		
Item	**Size**	**Cost**
Wax	500 g	\$16
	1 kg	\$26
Fragrance oil	30 ml	\$12
	60 ml	\$22
Candle wick	10 pieces, each 15 cm long	\$8

(a) Wei makes a candle using a mould in the shape of a cube with side length 5 cm.
Calculate the total mass of the candle.

Answer .... g [2]

(b) The total mass of another candle is 150 g.
Show that the mass of wax used for this candle is 139 g.

Answer

[2]

(c) Wei makes two sizes of candle using cylindrical moulds.

The cylinders are geometrically similar with diameters of 4 cm and 6 cm.

The height of the smaller cylinder is 8 cm.

Wei wants to make a small profit when he sells these candles.

Suggest a suitable amount for him to charge for each size of candle.

Justify the decisions you make and show your calculations clearly.

....

.... [7]